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Time–frequency ensemble network for wind turbine mechanical fault diagnosis

Haiyu Guo [a], Xingzheng Guo [a], Xiaoguang Zhang [b,c,d,*], Fanfan Lu [b], Chuang Liang [e]

[a] School of Electrical Engineering, Shenyang University of Technology, Shenyang, China
[b] Shanghai Intelligent Quality Technology Co., Ltd., Shanghai, China
[c] School of Computer Science and Technology, University of Science and Technology of China, Hefei, China
[d] Yangtze Delta Information Intelligence Innovation Research Institute, Wuhu, China
[e] School of Mechanical Engineering, Shenyang Ligong University, Shenyang, China

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ABSTRACT

Wind turbines typically operate under variable speed conditions, so the collected vibration signals are affected by non-linearity and information mixing, while also containing a large amount of noise interference. However, most existing methods extract fault features from a single domain, failing to capture the signals' diverse and complex characteristics. To fully exploit multi-domain discriminative features under variable speed conditions, this paper proposes a time–frequency ensemble network (TFNet). First, the feature representation is improved by constructing an adaptive spectral block (ASB) using Fourier analysis, while an adaptive threshold is introduced to reduce noise interference. Second, the Transformer and Graph Convolutional Network (GCN) are combined to extract the time–frequency discriminative features of defects. Specifically. In the time domain module, the global time domain features of faults are extracted by the Transformer encoder block. In the frequency domain module, a mixhop graph convolutional network is used to extract the multi-scale frequency domain features of different neighbours, and a Multi Head Attention (MHA) mechanism is introduced to capture the intra-feature dependencies. To achieve better diagnostic results under variable speed conditions, a label smoothing algorithm is used to assist the training of the model. A case study is conducted using the WT-Planetary gearbox dataset and the XJTUSuprgear variable speed gearbox dataset as well as the CWRU Bearing dataset. The experimental results show that the proposed model has high diagnostic accuracy and strong generalisation ability compared to other fault diagnosis models.

1. Introduction

Wind turbine gearboxes usually operate under complex, variable and harsh working conditions, and are very susceptible to failures, which can lead to failures will cause huge economic losses and even personal injuries [1]. Therefore, in order to ensure the normal operation of the equipment and reduce the cost of production and operation of enterprises, it is of great significance to carry out condition monitoring and fault diagnosis research on wind turbine gearboxes [2].

In recent years, the rapid development of big data and artificial intelligence has resulted in the growing application of machine learning in the field of fault diagnosis [3]. Tang et al. [4] proposed an improved butterfly optimisation algorithm to optimise the extreme random forest detection model for solving the difficulty of optimising the parameters of the gearbox detection model, achieving a dynamic balance between the global search and local search; In order to solve the nonlinear impulse characteristics of planetary gearboxes under strong noise, the literature [5] uses an improved genetic algorithm (IGA) to optimise the FMD parameters and construct a support vector machine model (SVM) for feature extraction; For the problem of difficult classification of compound faults in gearboxes, in the literature [6], Empirical Modal Decomposition (EMD) is used to decompose the vibration signals, and the multi-scale fuzzy entropy mean value of the preferred Intrinsic Mode Function (IMF) components is used as a feature vector, which is fed into Artificial Neural Networks (ANN) to obtain the optimal fault diagnosis and classification results; Liu et al. [7] implemented fault diagnosis of gearboxes by inputting the extracted and processed vibration signal features into an SVM classification model optimised by the Particle Swarm Optimisation (PSO) algorithm;

Traditional machine learning methods rely heavily on manual feature extraction by experts, which may not fully capture fault characteristics and thus compromise diagnosis accuracy [8].

* Corresponding author at: Shanghai Intelligent Quality Technology Co., Ltd., Shanghai, China.
E-mail address: zhangxiaoguang@ustc.edu.cn (X. Zhang).

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As an important branch of machine learning, deep learning (DL) effectively overcomes the limitations of traditional machine learning techniques by integrating feature extraction and fault diagnosis through an end-to-end architecture [9]. Some representative networks include Stacked Self-Encoder (SAE), Convolutional Neural Networks (CNN), Recurrent Neural Networks (RNN), and Long Short-Term Memory Networks (LSTM). Yin et al. [10] used multi-scale residual features (MRFs) to enrich fault information and proposed that an efficient channel attention (ECA)-stacked residual neural network can improve the accuracy of transmission fault diagnosis. Wang et al. [11] proposed a lightweight multiscale convolutional neural network (LMSCNN) for noise reduction of vibration signals using large kernel convolution to address the problems of limited samples and noise interference. Depth-separable convolution is used to mine fault features at different scales, and the effectiveness of the proposed method is verified on two different gearbox datasets. Yang et al. [12] proposed a practical and effective Self-Attention Parallel Fusion Network (SAPFN) that uses self-attention to optimally assign the weights of fault features and redundant features. The aim is to efficiently extract fault information from wind turbine gearbox signals. Graph structure data is an information-rich, non-Euclidean structure that contains node features and connection relationships. Chen et al. [13] proposed a novel explicit speed-integrated LSTM (SI-LSTM) model to improve the representation accuracy of non-stationary vibration signals and enhance the gearbox fault detection capability by considering account the effect of speed variation on vibration signals. Zhao et al. [14] developed a data-driven time–frequency analysis (TFA) technique, CTNet, by fusing a fully convolutional self-encoder network with a convolutional block attention module (CBAM). The instantaneous frequency (IF) of non-stationary fault signals of wind turbine planetary gearboxes and bearings with closely spaced transitions was addressed. Graph Neural Networks have great potential in fault diagnosis as deep learning algorithms that can learn graph data [15]. Wei et al. [16] proposed a transmission fault diagnosis method based on fine topology and spatio-temporal GCN. By combining GCN and GRU, a network model with spatio-temporal feature co-extraction is constructed to achieve multi-dimensional mining of fault information. Yin et al. [17] proposed a multi-scale graph convolutional network (MS-GCN) to identify potential relationships between signals at different scales. The multiscale feature extraction module and the multiscale graph iteration module, as well as the Bayesian fusion module, are mainly used to extract and synthesise the features of signals at different scales. Zhao et al. [18] proposed a Multi-Perceptual Graph Convolutional Tree Embedded Network (MPGCTN) to address the problem of imbalance in aircraft bearing data. A dual-channel feature map construction method is designed to convert the high-dimensional mapping into feature distance and dynamic maps to enhance the diversity of error information. It shows excellent accuracy under two aircraft engine bearing imbalance data sets.

Transformer [19]self-attention network has been rapidly and widely used in the field of natural language and machine vision due to the advantages of parallel computation, capturing long-distance dependency, and strong global feature learning ability, etc. At the same time, the research on fault diagnosis based on Transformer has become a hot topic for scholars at home and abroad [20]. Guan et al. [21] propose an optimised Transformer network, which effectively improves the capture of local feature information by introducing a multi-scale perceptual layer and linear embedding technique. The network is able to deeply analyse the fault features in multiple time dimensions, while accurately retaining the positional information of the original signals. Han et al. [22] propose a novel framework of Convformer-NSE for the analysis of different transmission systems to solve the problem that the local features of CNN cannot obtain global features. The Convformer is constructed to improve the non-linear representation of the feature map, and the long-range and local features are extracted using sparse modified multi-self attention. Faced with the difficulty of accurately identifying defects with few defect samples, Zhang et al. [23] proposed the multi-scale dilation convolution and swin-transformer (MSDC-Swin-T) method. The Coordinate Reconstruction Attention Mechanism (CRAM) is designed to improve the information acquisition; the multi-scale convolutional marker embedding is constructed to extract the local features, and the Swin-Transformer is used to capture the global dependencies and dig out the subtle fault features. Xiao et al. [24] proposed a Bayesian variational transformer (Bayesformer) in which all attention weights are considered as potential random variables to improve the generalisation of the Transformer.

An interesting progress on deep learning is the interpretation of network architecture and learned feature to break through the ‘black box’ nature of deep learning (DL-based) methods [25]. Recently, a model-driven deep unrolling method [26] has been proposed to achieve prior interpretability, whose core lies in unrolling the optimisation algorithm corresponding to a predefined model into a neural network. This network inherently possesses interpretability and demonstrates robustness against noise attacks. The effectiveness of the proposed model has been validated through two distinct gear fault experiments. An et al. [27] propose an interpretable neural network with a well-defined theoretical basis for the network architecture, which is also unrolling algorithmically to provide high-performance and credible mechanical fault diagnosis results. This novel algorithmic unrolling is also used in an adversarial algorithmic unrolling network [28] for interpretable mechanical anomaly detection and cleverly obtains interim and post hoc interpretations. These techniques have significantly advanced the development of fault diagnosis techniques.

Under stable conditions, the above-mentioned methods can yield favourable diagnostic outcomes. However, wind turbine frequently operates under varying conditions, such as variable speed and load variations, which makes the fault-related pulse information in the mechanical signal susceptible to noise suppression. The mechanical vibration signals are in essence complex 1-D time series. The collected data displays instability, and weak fault characteristics due to the complicated transmission path. Thus, under the strong noise interference of variable speed conditions, it is difficult to extract fault characteristics. This paper develops a time–frequency ensemble network (TFNet) to address this issue. First, an adaptive spectral block (ASB) constructed by means of Fourier analysis is utilised to reduce the interference of noise. Then, a time-domain Transformer encoder module (T-block) and a frequency-domain Mixhop GCN-MHA module (F-block) are constructed to effectively extract and integrate the multi-dimensional features of the vibration signals in both the time domain and the frequency domain, which achieves multi-domain feature complementarity, thereby enabling the enhancement of weak fault-related features. In addition, a label smoothing strategy is employed to assist model training, thereby enhancing the diagnostic performance under nonstationary operating conditions.

The main contributions of this paper are summarised below:

(1) In order to extract effective fault discrimination features, a novel feature extraction method is designed. It mainly consists of Transformer Encoding Blocks and Mixhop GCN-MHA, which are designed to extract time–frequency features at different scales.

(2) To obtain good diagnostic results under variable speed conditions, we use a label smoothing algorithm to help train the model.

(3) Using a wind turbine gearbox and bearing data set, the proposed TFNet is compared with other diagnostic methods to verify its effectiveness and superiority. Furthermore, using engineering cases, it is demonstrated that TFNet has significant advantages in noisy environments.

The rest of the paper is structured as follows. Section 2 presents the theoretical foundations of MHA and Transformer Encoder. Section 3 provides a detailed illustration of the designed TFNet. Sections 4 and 5 present the experimental setup, results and discussion under different data sets. Section 6 is dedicated to the technical validation. Finally, Section 7 provides a summary of the work presented.

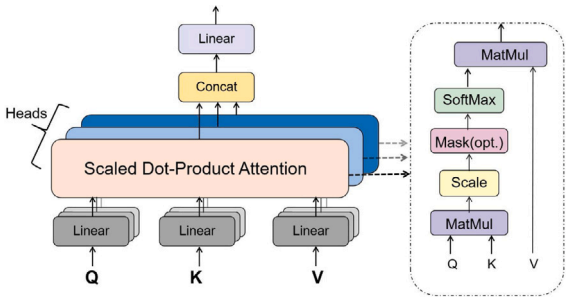


Fig. 1. MHA mechanism.

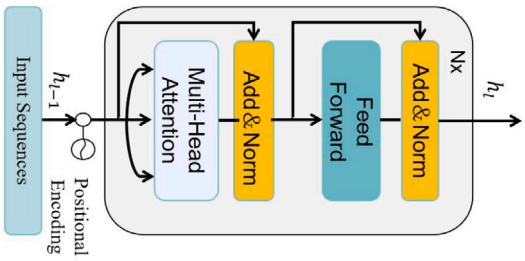


Fig. 2. Transformer encoder.

2. Theory background

2.1. Self-attention mechanism

The self-attention mechanism [29] can be regarded as a non-local filtering operation. Based on the estimation of the attention scores of each position, it calculates the response of each position of the sequence, and then collects the corresponding inputs according to the scores to extract features, grasp the association of elements, and improve the performance of the model. As shown in Fig. 1 (right), for input $X = \left[x_1, x_2, \ldots, x_n\right] \in \mathbf{R}^{n\times d}$, the final output sequence is $Y = [y_1, y_2, \ldots, y_n] \in \mathbf{R}^{n\times d}$, which is calculated as follows:

$$\boldsymbol{Y} = S_{\mathrm{A}}(\boldsymbol{Q}, \boldsymbol{K}, \boldsymbol{V}) = \mathrm{Softmax}(\boldsymbol{A}) \cdot \boldsymbol{V} = \mathrm{Softmax}\left(\frac{\boldsymbol{Q} \cdot \boldsymbol{K}^{\mathrm{T}}}{\sqrt{d}}\right) \cdot \boldsymbol{V} \tag{1}$$

$$\begin{cases} Q = W_q \cdot \boldsymbol{X} \\ K = W_k \cdot \boldsymbol{X} \\ V = W_v \cdot \boldsymbol{X} \end{cases} \tag{2}$$

where: n is the sequence length; d is the number of dimensions; Q, K, V are the three intermediate matrices; W_q, W_k, W_v are three different linear transformation matrices, $W_q, W_k, W_v \in \mathbf{R}^{d\times d}$; $S_A(\cdot)$ is the self-attentional computation; A is the self-attentional matrix, $A \in \mathbf{R}^{n\times n}$, where the individual elements denote the attentional values between two elements in X.

2.2. Multi-head attention mechanism

The MHA mechanism is composed of multiple self-attention mechanisms spliced together, which can avoid the problems that the self-attention mechanism may have, such as information omission or excessive focus on certain local information. As shown in Fig. 1 (left), when the number of heads is set to h, this requires the application of h separate linear transformation matrices, through which three sets of intermediate matrices can be obtained: $\{\boldsymbol{Q}_i\}_{i=1}^{h}$, $\{\boldsymbol{K}_i\}_{i=1}^{h}$, and $\{\boldsymbol{V}_i\}_{i=1}^{h}$. The procedure is described below:

$$\begin{cases} S_{\mathrm{MHA}}(\boldsymbol{Q}', \boldsymbol{K}', \boldsymbol{V}') = \mathrm{concat}\left(\boldsymbol{H}_1, \ldots, \boldsymbol{H}_h\right) \cdot W \\ \boldsymbol{H}_i = S_{\mathrm{A}}\left(\boldsymbol{Q}_i, \boldsymbol{K}_i, \boldsymbol{V}_i\right); i = 1, 2, \ldots, h \end{cases} \tag{3}$$

where: Q', K', V' are composed of Q_i, K_i, V_i splices respectively; W is the linear transformation matrix, $W \in \mathbf{R}^{d\times d}$; $concat(\cdot)$ is the splicing operation.

2.3. Transformer encoder

The Transformer encoder [30] incorporates positional encoding, multi-head attention, a feed-forward network, and residual layer normalisation (Fig. 2). The multi-attention mechanism assigns weights to the input feature sequences to focus on key information, the feed-forward network maps the multi-attention output to a non-linear space, and the residual and layer normalisation layers are used to mitigate gradient vanishing, accelerate convergence, and improve generalisation.

(1) Positional Encoding. Since the self-attention mechanism does not take into account the positional relationships of the elements in the sequence, the inclusion of sequence positional information through positional coding is achieved by alternating between the even and odd dimensions of the input feature samples through the sin and cos functions. As shown in the following equation:

$$\begin{cases} \boldsymbol{PE}_{(pos,2i)} = \sin(pos/10000^{2i/d_{model}}) \\ \boldsymbol{PE}_{(pos,2i+1)} = \cos(pos/10000^{2i/d_{model}}) \end{cases} \tag{4}$$

where pos is the position index in the sequence. i is dimension index that distinguishes coded dimensions. d_{model} is the dimension of the model, i.e. the length of the encoding vector for each position. $PE_{(pos,2i)}$ is the 2ith element of the coding vector for position pos, generated by the sine function. $PE_{(pos,2i+1)}$ denotes the 2i+1st element of the encoding vector for position pos, generated using the cosine function. The $10000^{2i/d_{model}}$ is a frequency term to control the frequency variation in different dimensions.

(2) Residual Connection. In the Transformer model, residual connections help address the issues of vanishing gradients and the degradation of weight matrices. When combined with Layer Normalisation, the residual connection can further enhance the training stability of the model:

$$R_{\mathrm{C}} = L_{\mathrm{N}}\left[X + S_{\mathrm{A}}(X)\right] \tag{5}$$

where: $R_{\mathrm{C}}(\cdot)$ is the residual concatenation operation; $L_{\mathrm{N}}(\cdot)$ is the layer normalisation operation; $S_{\mathrm{MHA}}(\cdot)$ is used instead of $S_A(\cdot)$ when using multi-head attention.

(3) Feedforward Network. The feedforward network in Transformer consists of two linear transformation layers and one nonlinear activation layer, the principle is expressed as follows:

$$N_{\mathrm{FFN}}(X) = W_2 \cdot f(W_1 \cdot X) \tag{6}$$

In this context, $N_{FFN}(\cdot)$ represents the feed-forward network. Here, W_1 and W_2 stand for the values associated with the two individual linear layers respectively. Meanwhile, $f(\cdot)$ denotes a nonlinear activation function, with an example being the GELU function.

3. Proposed method

When a single time-domain feature is used for diagnosis, it provides only partial information about the fault to the model. To accurately assess the severity of the fault, the various characteristics of the fault are taken into account. The flow of the proposed multi-feature fusion TFNet for gearbox fault diagnosis model is shown in Fig. 3 with the following steps:

Step1: The fault data were gathered and transmitted via a signal acquisition system. Subsequently, the dataset was partitioned into training, validation, and test sets in the proportion of 6:2:2.

Step2: Reducing Noise Interference by Building Adaptive Spectrum Blocks;

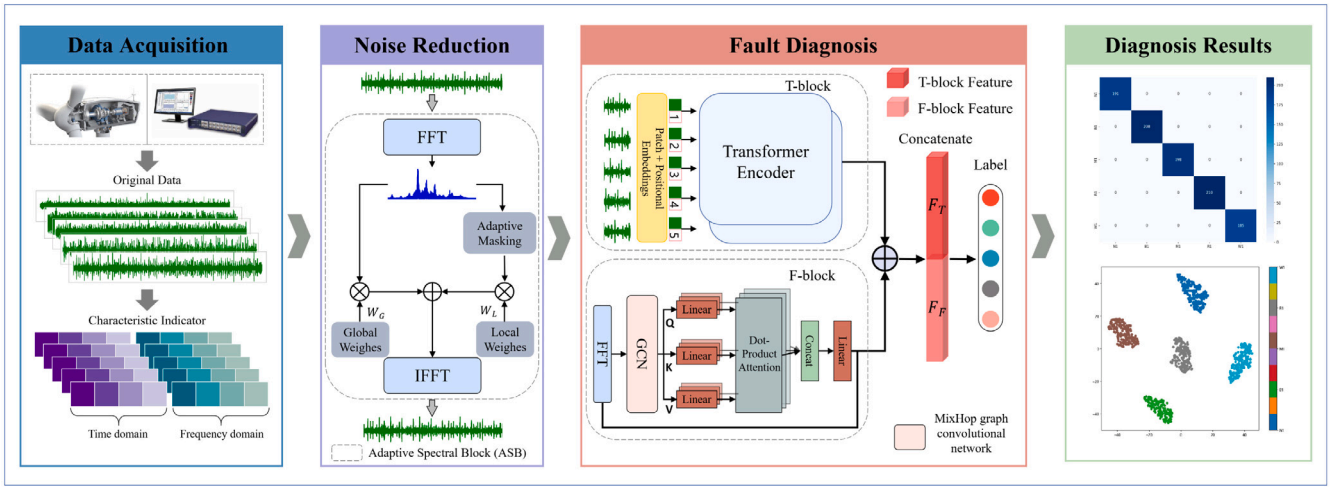


Fig. 3. The flow chart of the proposed approach to fault diagnosis.

Step3: Build a TFNet network model, randomly initialise the weights and bias terms of this model, and set the appropriate hyperparameters;

Step4: Use validation sets to verify the effectiveness of model training and optimise model parameters based on the validation results;

Step5: The performance of the trained TFNet network model is tested to calculate the fault diagnosis accuracy and output the fault classification results.

3.1. Adaptive spectral block

Specifically, the Adaptive Spectrum Block [31](ASB) uses Fourier analysis to optimise the representation of features to effectively capture long- and short-term interactions in the data. At the same time, the adaptive thresholding strategy reduces noise interference, enhancing data processing accuracy and reliability for subsequent analysis.

For a given discrete time series $x[n]$, its frequency domain representation $X[k]$ is obtained by performing a Fast Fourier Transform (FFT) along the spatial dimension, which can be characterised exactly by Eq. (7). Similarly, for a given S_V, the corresponding representation is computed as described below:

$$F = \mathcal{F}[S_V] \in \mathbb{C}^{C\times L'} \tag{7}$$

In the above equation, the symbol $F[\cdot]$ represents the one-dimensional Fast Fourier Transform (FFT) operation. Where L' refers to the value of the length of the transform sequence in the frequency domain. Note that the value of L' may not be the same as the original L due to differences in the implementation of the FFT and the characteristics of the time series data.

After acquiring the frequency domain of time series data representation F by Fast Fourier Transform (FFT), the first step is to calculate the power spectrum of F. This process is crucial for accurately identifying the major frequency components of the data and is centred on the squaring of the amplitude of the frequency components. Specifically, the formula for the power spectrum P can be written explicitly as $\boldsymbol{P} = |\boldsymbol{F}|^2$. The resulting power spectrum P can effectively quantify the intensity distribution of time series data at different frequencies, providing an important metric for in-depth analysis of the frequency characteristics of the data to more fully and accurately grasp the intrinsic frequency structure of the data information.

The core of effective noise reduction is the adaptive filtering of the high frequency components of the power spectrum P. This is achieved by means of a trainable threshold θ, which in the training phase is set to a learnable parameter that can be optimised by backpropagation and is expressed as $\frac{\partial \mathcal{L}}{\partial \theta}$. In this way, the threshold θ is able to accurately distinguish the fundamental signal frequency from the noise component. The corresponding adaptive threshold formula is shown below:

$$F_{\text{filtered}} = F \odot P \tag{8}$$

where the operator $\odot$ represents the arithmetic rule for multiplying by an element. The operator $(P > \theta)$ is used here as a binary mask, which means that if the power value of a frequency component in the power spectrum P is greater than the threshold θ, the corresponding position in the mask is 1, which means that the frequency is retained; and when the power value of the frequency component is less than or equal to the threshold θ, the corresponding position in the mask is 0, which means that the frequency is filtered out, thus achieving selective retention or removal of different frequency components in the power spectrum to achieve a specific noise reduction effect. This means that this frequency will be filtered out to achieve selective retention or removal of different frequency components in the power spectrum to achieve a specific noise reduction effect.

After the adaptive filtering operation on the frequency domain data, the model applies two sets of filters with learnable characteristics. The global filter is obtained by learning the original frequency domain data F, while the local filter is learned from the adaptively filtered data F_{filtered}. Let us assume that W_{G} represents the learnable global filter and W_{L} represents the learnable local filter. Then the specific way of using these two types of filters can be expressed as follows:

$$\begin{cases} F_{\text{G}} = W_{\text{G}} \odot F \\ F_{\text{L}} = W_{\text{L}} \odot F_{\text{filtered}} \end{cases} \tag{9}$$

Next, comprehensive feature details are captured by integrating these filtered features, i.e., $\boldsymbol{F}_{\textbf{integrated}} = \boldsymbol{F}_{\textbf{G}} + \boldsymbol{F}_{\textbf{L}}$

To convert the integrated frequency domain data back to the time domain, the Inverse Fast Fourier Transform (IFFT) technique is employed. The resultant time domain signal $\boldsymbol{S}'$ can be expressed through the following equation:

$$S' = \mathcal{F}^{-1}[F_{\text{integrated}}] \in \mathbb{R}^{C\times p'} \tag{10}$$

Improved features are maintained by IFFT to ensure that the enhanced features are aligned with the original data structure of the entered time series.

3.2. T-block

The T-block uses the Transformer encoder to capture global dependencies in the time domain. To better extract temporal global features, the Transformer coding layer also incorporates Positional Encoding.

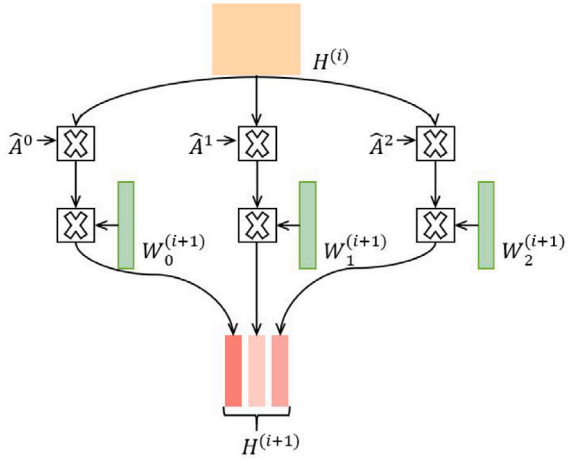


Fig. 4. MixHop graph convolutional layer.

It first divides the input sequence $X \in \mathbb{R}^L$ into a number of blocks $X_p \in \mathbb{R}^{P\times N}$, where P is the length of the blocks and N is the number of blocks. Time series patch X_p After processing by the multi-head attention mechanism and location coding, the feed-forward neural network (FFN) further integrates and transforms the features. The FFN non-linearly transforms the feature vectors output from the multi-head attention mechanism, which enhances the model's ability to represent global features in the time domain. Meanwhile, residual connectivity and layer normalisation mechanisms ensure efficient information transfer in the network and stable training. The final output $X_o^{(t)} \in \mathbb{R}^T$ is generated by linear projection.

3.3. F-block

The F-block uses a MixHop graph convolutional network [32] in conjunction with a multi-head attention mechanism to extract frequency domain features. After the MixHop network has extracted different order neighbour features, these features are fused using the multi-head attention mechanism. Depending on the attention weights of the different heads, the model is able to capture the correlation and importance between features more flexibly.

In Graph Convolutional Networks(GCNs), information propagates through the neighbourhood of a node. The basic principle of a traditional GCN is shown in Eq. (11):

$$\hat{H}^{(l+1)} = \sigma\left(\hat{D}^{-\frac{1}{2}}\hat{A}\hat{D}^{-\frac{1}{2}}H^{(l)}W^{(l)}\right) \tag{11}$$

where $H^{(l)}$ is the node feature matrix of layer l, $\hat{A} = A + I$ (A is the adjacency matrix, I is the unit matrix), $\hat{D}$ is the degree matrix of $\hat{A}$, σ is the activation function and $W^{(l)}$ is the weight matrix of layer l. However, traditional GCNs mainly focus on the first order neighbourhood information of the nodes.

The core aspect of Mixhop graph convolutional networks lies in their proficiency in integrating neighbourhood information of varying orders. Illustrated in Fig. 4, suppose $X \in R^{n\times d}$ represents the input node feature matrix, where n indicates the quantity of nodes and d signifies the feature dimension. At the same time, assume $A \in R^{n\times n}$ to be the adjacency matrix. On this foundation, the formulation of the Mixhop convolutional layer is given as below:

$$H^{(i+1)} = \|_{j\in}\sigma\left(\hat{A}^j H^{(i)} W_j^{(i)}\right) \tag{12}$$

Here $\hat{A}^j$ denotes the jth power of the adjacency matrix A, which is used to obtain the jth order neighbourhood information of the node. By manipulating the neighbourhood information of different orders, Mixhop is able to mix neighbourhood features of different distances and obtain $H_{Mixhop}^{(l)}$.

The MHA operation is then performed:

$$H_{Mixhop-MHA}^{(l)} = W^O\left(\|_{i=1}^h head_i\right) \tag{13}$$

Here $head_i = Attention(Q_i, K_i, V_i)$, $Q_i = H_{Mixhop}^{(l)} W_i^Q$, $K_i = H_{Mixhop}^{(l)} W_i^K$, $V_i = H_{Mixhop}^{(l)} W_i^V$.

Finally, the final frequency domain feature representation $H_{F-block}$ is obtained after residual linkage operations.

$$H_{F-block} = H_{Mixhop-MHA}^{(L)} + H^{(i)} \tag{14}$$

To preserve the respective feature information, the time and frequency domain features were integrated by direct concatenation along the feature dimensions.

$$F_{out} = Cat(H_{T-block}, H_{F-block}) \tag{15}$$

where F_{out} denotes the final output feature and $H_{T-block}$ $H_{F-block}$ are time and frequency domain features, respectively.

3.4. Label smoothing algorithmk

In this study, a label smoothing [33] strategy is introduced to optimise the model training process, which improves the diagnostic performance of TFNet under speed fluctuations and enables it to identify different types of faults more accurately, effectively enhancing the adaptability and reliability of the model under complex working conditions. Given a dataset $W_{train} = \{(x_i, y_i)\}$ (with N categories), x_i is the sample, y_i is the label, and the TFNet prediction is $p\left(n|x_i\right)$. The cross-entropy loss is calculated as follows:

$$L_{loss} = -\sum_{n=1}^{N} q\left(n|x_i\right) logp\left(n|x_i\right) = -logp\left(n = y_i|x_i\right) \tag{16}$$

where $p\left(n|x_i\right)$ is equal to 1 if the result belongs to category N and 0 if the result belongs to any other category.

In model training, the label smoothing method replaces hard labels with soft labels to smooth their distribution. Specifically, the improved distribution $q'\left(n|x_i\right)$ can be obtained as follows:

$$q'\left(n|x_i\right) = (1-k)\,q\left(n|x_i\right) + \frac{k}{N} \tag{17}$$

where k is the smoothing parameter, which routinely takes the value 0.1, this setting is used in model training to achieve a specific label smoothing effect to optimise model performance.

4. Fault identification of WT-planetary gearbox dataset

4.1. Data description and experimental settings

In this section, the WT-planetary gearbox data set collected from the planetary gearboxes of the wind power drive train testbed of Beijing Jiaotong University is used to verify the model diagnosticity [34]. The experimental setup is composed of a motor, a planetary gearbox, a stator gearbox, and a load cell, as illustrated in Fig. 5. Within the planetary gearbox, four planetary gears revolve around the sun gear, which can exhibit five distinct conditions: normal operation, breakage, wear, root cracking, and tooth loss, as shown in Fig. 6. The vibration data were acquired using a Sinocera CA-YD-1181 accelerometer, while the velocity pulses were acquired via an encoder. For all channels, the sampling rate was 48 kHz.

In the present dataset, for every single health state, there exist eight diverse operating conditions. In order to evaluate the model's generalisability, four distinct conditions, specifically 20 Hz, 30 Hz, 40 Hz, and 50 Hz, were selected for the experimental procedure. Under each health circumstance, each sample encompassed 1024 measurement points, and a total of 500 samples were collected. Concurrently, all of these samples were partitioned into the training set, test set, and validation set according to the proportion of 6:2:2. The data set is illustrated in Table 1.

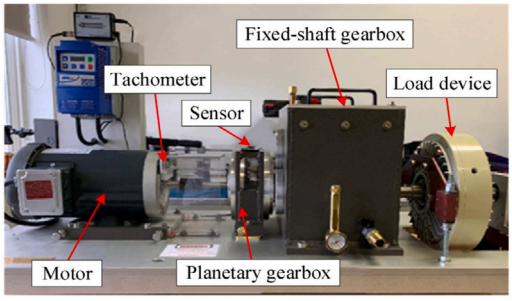


Fig. 5. Wind turbine drive train test rig.

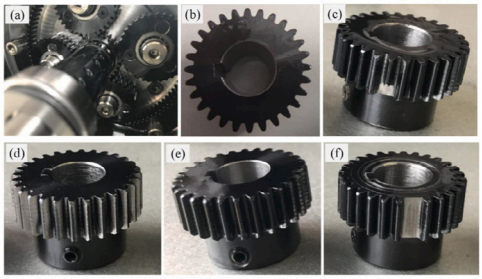


Fig. 6. (a) Internal structure of planetary gearbox. (b) healthy condition, (c) gear breakage, (d) wear gear, (e) root crack, and (f) missing tooth.

Table 1
The detailed information of WT-Planetary gearbox dataset.

Operating condition	Health condition	Sample size	Class labels
20 Hz/30 Hz 40 Hz/50 Hz	Healthy	500	N1
	Gear breakage	500	B1
	Wear gear	500	W1
	Root crack	500	R1
	Missing tooth	500	M1

To elaborate in detail on the relationship between the extracted time-domain and frequency-domain features and different fault types, the time-domain and frequency-domain signal diagrams of gears in different states are analysed, revealing their respective characteristics.

In the time-domain, as shown in Fig. 7, the signal of a healthy state features a regular waveform with stable amplitude changes. The signal of gear breakage has obvious impact peaks. The signal of gear wear shows large amplitude fluctuations. The signal of root crack is characterised by irregular jittering, and the signal of missing tooth has intermittent abnormal fluctuations.

In the frequency-domain, as depicted in Fig. 8, the signal of a healthy state has energy concentrated at specific frequency points. The signal of gear breakage has energy distributed across multiple frequency ranges and is relatively dispersed. The signal of gear wear has significantly enhanced energy in certain frequency bands. The signal of root crack presents multiple discrete peaks, and the signal of missing tooth has a wide energy distribution with no obvious pattern.

4.2. Hyperparameter selection

The hardware environment used for the simulation experiments in this paper is: Window 11 with 12 GB of RAM, equipped with RTX 4090 GPU for model training, PyCharm editor, Python version 3.9, and based on the Pytorch framework for design and validation.

TFNet network structure: the batch size is 32, the initial learning rate is 0.0001, the activation function is a Relu function, the Adam adaptive optimiser is used to update the network parameters to obtain the optimal value of the parameters, and the dropout value is chosen

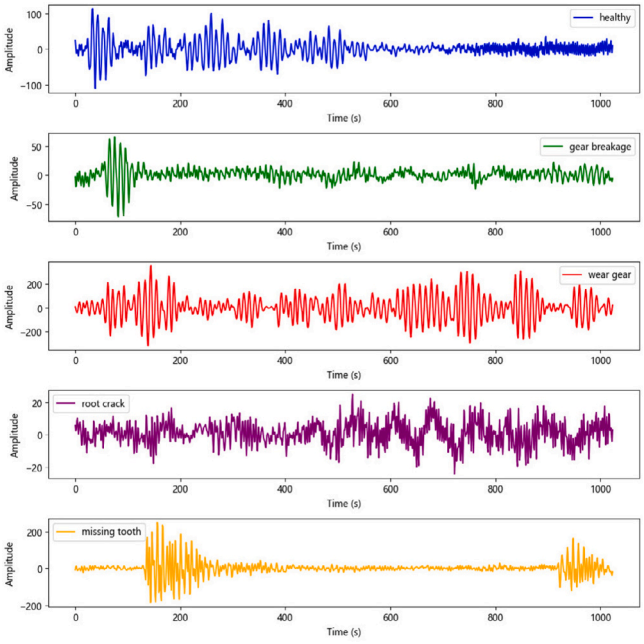


Fig. 7. Time domain signal diagram.

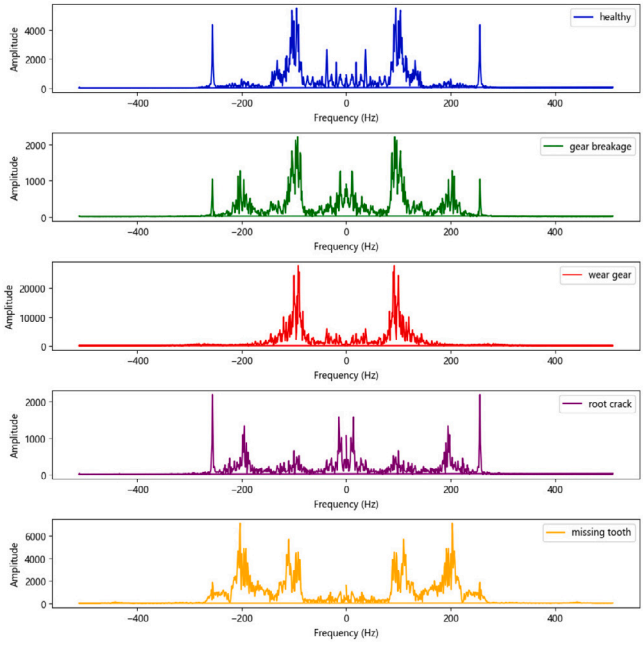


Fig. 8. Frequency domain signal diagram.

as 0.5. In addition, it should be noted that an inappropriate choice of hyperparameters not only reduces the mean accuracy, but also makes the network performance more unstable and the error distribution more scattered.

We will determine the model hyperparameter choices based on the validation of the WT-Planetary Gearbox dataset with an acquisition frequency of 20 HZ, and assess the impact of these hyperparameter choices on model size and generalisation performance. To improve the feasibility of the research work to perform multiple averaging validation and analysis. Hyperparameters to be determined include : (A) the number of T-Blocks N_1, (B) the number of attention heads h, (C) the embeddingd_e and hiding dimensionsd_h, and (D) the number of

Table 2
Model selection and influence of some hyperparameters.

Base	N_1	h	d_e	d_h	K_h	Average acc (%)	Training times (s)
	4	4	128	256	3	99.45	843
A	2					98.21	721
	6					99.16	965
	8					99.48	1233
B		1				95.75	674
		2				97.42	785
		8				99.55	995
		16				99.62	1175
C			16	32		98.16	639
			32	64		98.41	760
			64	128		99.03	833
			256	512		99.62	1056
D					1	97.64	726
					2	98.64	810
					5	99.77	968

Table 3
Comparison of evaluation indicators results for different operating conditions test sets.

Operating conditions	Accuracy (%)	Precision (%)	Recall (%)	F1_score (%)
20 Hz	99.95	99.95	99.95	99.95
30 Hz	99.52	99.54	99.50	99.50
40 Hz	99.28	99.32	99.32	99.28
50 Hz	98.88	98.83	98.79	98.79

different hops (k-hops) of MixHop K_h. Each model was trained using 5 validations. Specifically, the average of the validation results was used. The test results for each group are shown in Table 2, where base denotes the model finally selected for the subsequent study, average acc denotes the average test accuracy.

As can be seen from the table, the choice of these parameters has an impact on the training time and the performance of the model. In particular, the different dimensions and the number of T-blocks have a significant impact. If the dimensions of the embedding and hidden layers are too small, the network will not be sufficiently parameterised and the network performance will be poor. Conversely, if the dimensionality is too large, the network will be overparameterised and the training time will increase significantly.

4.3. Diagnostic results under different operating conditions

Fig. 9 shows the training/testing accuracy and loss curves after 100 iterations for four operating conditions with five sample signals. As can be seen in Fig. 9, the network model shows good learning ability and stability during the training process under different working conditions. As the number of iterations increases, the accuracy of the model on the test set reaches 99.98%, 99.96%, 99.90%, and 99.78%, respectively, and the loss rate is close to 0. Specifically, the model reaches a high accuracy at about the 5th iteration, and the fluctuation during the whole training process is small, and it tends to converge very quickly, which means that the model has an excellent learning speed and can capture the key features of the faults in a short time.

Since only accuracy and loss rate are used, it is difficult to show the superior characteristics of the proposed method in a comprehensive way. In order to obtain a more comprehensive and in-depth model performance, the model is further comprehensively analysed and evaluated using precision, recall, F1_score and other metrics, and the formula for each evaluation metric is [35]:

$$\text{Precision} = \frac{TP}{TP + FP} \tag{18}$$

$$\text{Recall} = \frac{TP}{TP + FN} \tag{19}$$

$$\text{F}_{1_score} = 2 \times \frac{Precision \times Recall}{Precision + Recall} \tag{20}$$

where Precision is a function of accurately classified cases (True Positive) and incorrectly classified positive cases (False Positive), Recall is a function of incorrectly classified cases (False Negative), and F1_score is the weighted average of Precision and Recall. Table 3 shows in detail the experimental results of each evaluation metric on the test set under different operating conditions after 100 epochs of training iterations.

In addition, the model performance was evaluated using a confusion matrix, and Fig. 10 shows the results of fault type identification for the model under four operating conditions. Fig. 10 indicates that under the 40 Hz condition, individual errors occur in the prediction of each fault type; under the 50 Hz condition, the main occurrence of errors in the prediction of gear breakage is mainly due to the fact that the vibration conditions of the data segment are similar to the normal state, and the individual samples are predicted to be in the normal state; under the 20 and 30 Hz conditions, there are no errors in the fault diagnosis, which indicates that the proposed method has an excellent fault identification and classification ability of the proposed method.

4.4. Comparison of different diagnostic methods

To demonstrate the advantage of the proposed method in planetary gearbox fault diagnosis, we compare it with seven popular deep learning networks. To ensure the fairness of the comparison test, the test is conducted at 20 Hz conditions and the training parameters of the comparison models are consistent with the proposed method.

(1) CNN [36]: Convolutional neural network is a common convolutional model used for fault identification tasks.

(2) WDCNN [37]: WDCNN uses different convolutional kernels to assign different weights to channels or feature maps to better focus on important features.

(3) Resnet [38]: Resnet is designed to allow the network to be continuously enhanced with increasing depth by introducing residual connections, thus extracting more discriminative and representative features.

(4) LSTM [39]: LSTM can efficiently handle long sequence data by modelling long-term dependencies in the sequence to better capture sequence information.

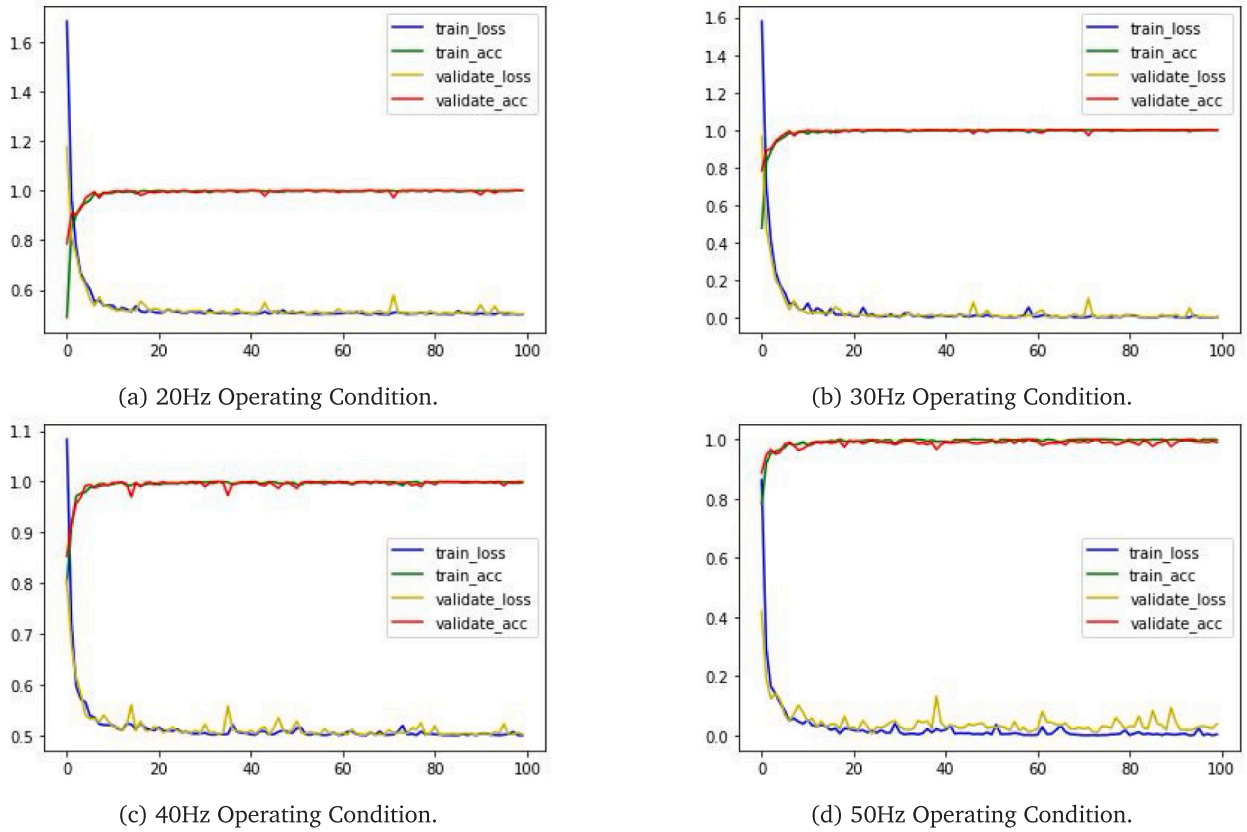
(5) MLFNet [40]: MLFNet is a multilayer feature fusion network based on the CNN model for multi-scale feature extraction and multilevel feature fusion of vibration signals.

(6) MSACNN [41]: MSACNN is based on Multiscale Attention Convolutional Neural Network, which introduces the multiscale attention mechanism into a one-dimensional convolutional neural network to obtain more important feature information.

(7) MSECTN [42]: MSECTN is based on the Multiscale Enhanced Convolutional Transformer Network, which performs multiscale convolutional embedding of kernels of different sizes to enrich the feature representation.

Table 4 shows the experimental results of different models on the test set after 100 training iterations for each evaluation metric. Comparison of the running time of each model is also referenced [43]. The table shows that the TFNet model achieves 99.95% in terms of accuracy, precision, recall and F1 score, and the training time is 843 s. Compared to other traditional models such as CNN, WDCNN, etc., TFNet has significant advantages in diagnostic performance, although the training time is slightly longer, but the overall performance is better.

The experiments were repeated five times for each method to circumvent chance, and the results are shown in Fig. 11. It is clear that the proposed TFNet diagnostic method is the most effective, with the accuracy of the results of the five tests being 99.94%, 99.95%, 99.92%, 99.88%, and 99.96%, respectively. The optimal accuracy of the TFNet is superior to that of the best of the other five methods. In addition, the standard deviation of TFNet is the smallest and the model performance is more stable compared to the other methods. Therefore, it can be concluded that TFNet is more effective than the other seven methods.



(a) 20Hz Operating Condition.

(b) 30Hz Operating Condition.

(c) 40Hz Operating Condition.

(d) 50Hz Operating Condition.

Fig. 9. Plot of accuracy and loss rate versus the number of training rounds of the model under different working conditions.

Table 4
Comparison of evaluation indicators results for different model test sets.

Model	Accuracy (%)	Precision (%)	Recall (%)	F1_score (%)	Training times (s)
TFNet	99.95	99.95	99.95	99.95	843
CNN	93.31	93.35	93.40	93.31	352
WDCNN	95.75	95.88	95.85	95.75	396
Resnet	96.27	96.25	96.25	96.22	783
LSTM	97.66	97.86	97.66	97.67	452
MLFNet	98.55	98.45	98.45	98.45	916
MSACNN	98.86	98.33	99.02	97.97	1159
MSECTN	99.31	98.90	99.56	99.13	1327

In the further analysis of the diagnostic results, in order to assure that the precision of the experimental model in terms of fault classification and the effectiveness of the fault data processing are fully verified, the features are discretised using the t-SNE method in order to provide more easily understandable results. Fig. 12 shows that TFNet outperforms the other methods in recognising changes between categories. It is worth noting that the feature maps generated by TFNet show that familiar categories are grouped together without much overlap between different categories.

4.5. Anti-noise performance test

Considering the complex mechanical environment and external factors in the actual operation of wind turbine gearboxes, the collected vibration signals will inevitably be contaminated by environmental noise. Now, by adding different Gaussian white noise to the vibration signals collected under 20 Hz conditions to simulate the signal quality of the planetary gearbox under actual working conditions, we aim to investigate the proposed method and the performance of each model under noise interference, and the experimental results are shown in Fig. 13. The SNR is calculated from the signal power and noise power using the following formula:

$$SNR_{dB} = 10\lg_{10}\left(\frac{P_{signal}}{P_{noise}}\right) \tag{21}$$

Based on the original vibration signals, a test data set with SNRs in the range of −10 to 6 dB is constructed by Eq. (21) to compare the effectiveness of different methods in various noise scenarios. The corresponding experimental results are presented in Fig. 13. It is clear that noise exerts a significant influence on the performance of each method. In particular, the diagnostic accuracy of these models shows a clear trend of decreasing with the gradual increase of noise intensity. Fig. 13 shows that compared with the other five models, the proposed method can still maintain a relatively high diagnostic accuracy even under the unfavourable conditions of noise interference. This indicates that the introduction of the adaptive spectrum block helps the model to focus on key features more accurately, which effectively reduces the interference caused by noise and improves the robustness and reliability of the model in the complex noise environment.

4.6. Ablation experiment

The proposed TFNet focuses on using Transformer and MixHop GCN to explore the multi-domain feature information obtained from

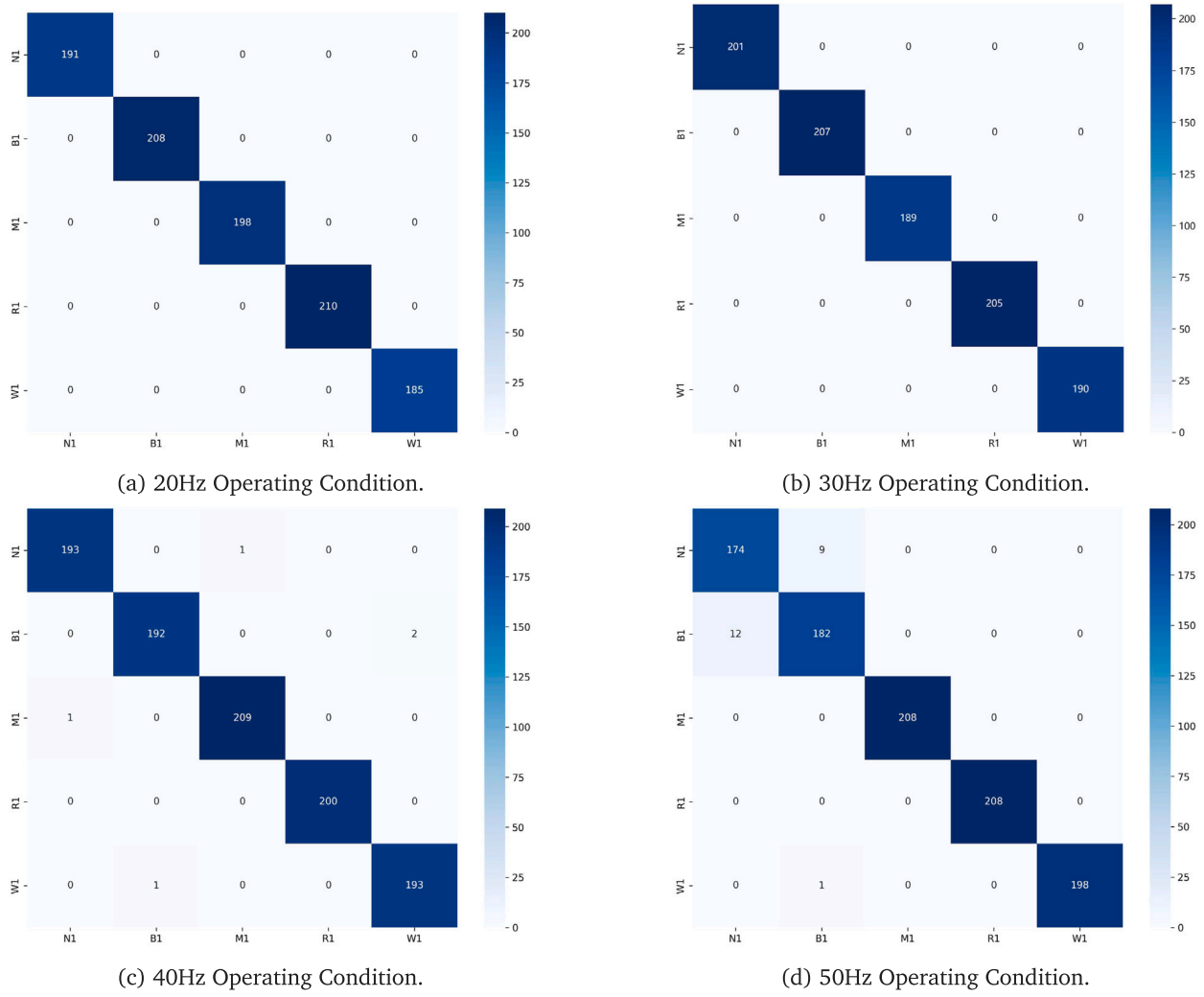


Fig. 10. Comparison of confusion matrices for each operating condition.

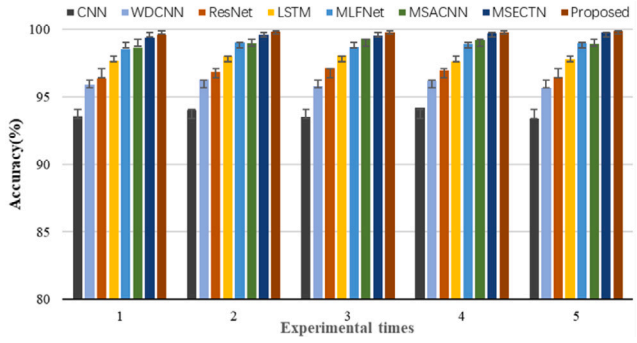
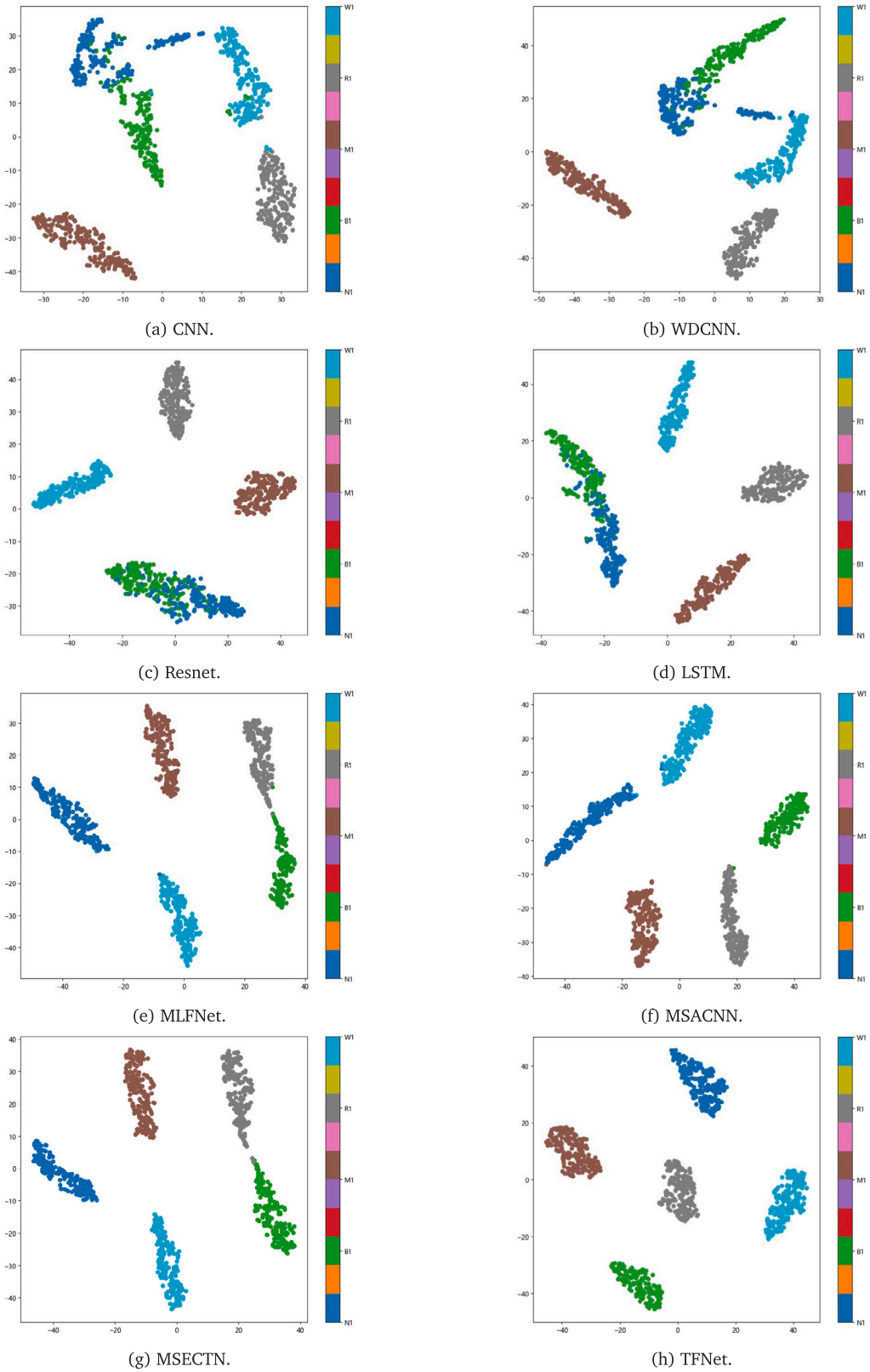


Fig. 11. Results of different methods for WT-Planetary gearbox dataset.

vibration signals. To evaluate the effectiveness of the different modules, a series of ablation experiments are performed, including single-domain features as well as different learning modules and multi-domain feature fusion methods.

We constructed four architectures, TFNet-nASB, TFNet-n(F-Block), TFNet-n(T-Block), and TFNet-nMHA, for the comparison task. TFNet-nASB, TFNet-n(F-Block), TFNet-n(T-Block), and TFNet-nMHA, have similar architectures to TFNet. TFNet-nASB and TFNet are different from each other because TFNet-nASB does not have a ASB. Similarly, TFNet-nMHA does not have a MHA. TFNet-n(F-Block) and TFNet-n(T-Block) are distinct from TFNet by using a single T-Block and F-Block, respectively. The WT-Planetary gearbox dataset with signal-to-noise ratios of −6, 0 and 6 dB at 20 Hz operating conditions is used for experimental validation. The results of the ablation experiments with different S/N ratios are given in Table 5.

We can see that the performance of TFNet is 0.77%, 1.47% and 0.95% better than TFNet-n(F-Block) and 1.63%, 1.57% and 0.81% better than TFNet-n(F-Block) and TFNet-n(T-Block) in the three cases, respectively. This is due to the fact that extracting only a single time–frequency feature cannot represent the fault characteristics comprehensively. In addition, TFNet performs 1.16%, 2.08% and 1.42% better than TFNet-nMHA in the three cases, respectively. Meanwhile, it is observed that TFNet far better than TFNet-nASB algorithm in all three scenarios. Especially in the strong noise condition with SNR −6 dB, TFNet outperforms TFNet-nASB by 3.11%. In fact, ASB can reduce noise interference by Fourier analysis of enhanced features and local filtering, which would improve the performance of TFNet in noisy conditions (see Table 5).



(a) CNN. (b) WDCNN.

(c) Resnet. (d) LSTM.

(e) MLFNet. (f) MSACNN.

(g) MSECTN. (h) TFNet.

Fig. 12. T-SNE feature visualisation for features under different models.

Table 5
Experimental results of ablation experiments under different noise conditions.

Model	SNR = −6 dB	SNR = 0 dB	SNR = 6 dB	Traning time (s)
TFNet	94.42%	99.95%	99.99%	843
TFNet-nASB	91.31%	97.64%	98.82%	815
TFNet-n(F-Block)	93.65%	98.48%	99.04%	417
TFNet-n(T-Block)	92.79%	98.38%	99.18%	532
TFNet-nMHA	93.26%	97.87%	98.57%	753

Table 6
The detailed information of XJTUSuprgear gearbox dataset.

Operating condition	Health condition	Sample size	Class labels
0~1200~0 (rmp)	Healthy	500	N
	Crack(0.2 mm)	500	C02
	Crack(0.6 mm)	500	C06
	Crack(1.0 mm)	500	C10
	Crack(1.4 mm)	500	C14

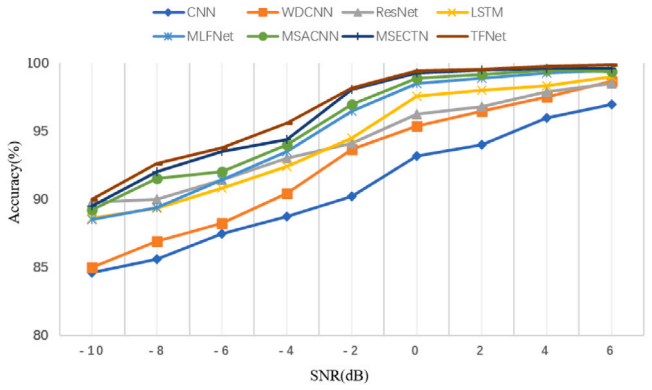


Fig. 13. Diagnostic accuracy of different methods using WT-Planetary gearbox dataset under different SNR conditions.

5. Fault identification of XJTUSuprgear gearbox dataset and CWRU bearing dataset

5.1. Fault identification of XJTUSuprgear gearbox dataset

In this section, the XJTUSuprgear gearbox dataset [44] is used with the aim of validating the diagnostic performance of the model under variable speed operating conditions. Specifically, the rotational speed varies from 0 to 1200 revolutions per minute (rpm) and then returns to 0. Such variable-speed conditions are more similar to the complex working conditions that gearboxes may encounter in practical engineering applications. The relevant experimental setup is illustrated in Fig. 14. This platform is mainly composed of a drive motor, a belt, a shaft, and a gearbox. The drive motor employed here is an AC inverter motor, with its power supplied by single-phase alternating current (220 V, 60/50 Hz). In the experimental process, four kinds of cracks of different degrees were carefully processed on the spur gear, details are shown in Fig. 15, and five kinds of vibration signals were collected in combination with the health status. The sampling frequency was 10 kHz throughout the experiment, and for each type of fault, 500 data were used, which were divided into training, validation, and test sets according to the ratio of 6:2:2. The details of the experimental data sets are shown in Table 6.

The training/testing accuracy and loss plots after 100 iterations are shown in Fig. 16. During training, both the training loss and validation loss exhibit rapid decline followed by stabilization with minimal fluctuations at low values, while the training accuracy and validation accuracy maintain consistently high levels with little variation. This demonstrates that the model converges effectively, shows progressive enhancement in its fitting capacity, delivers stable performance across

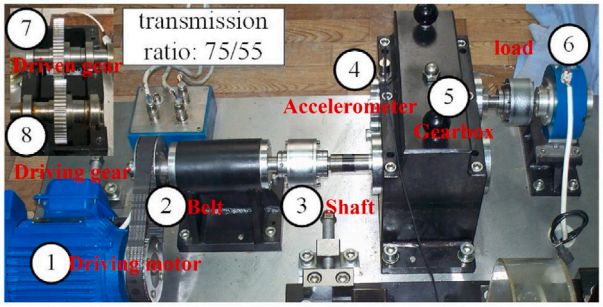


Fig. 14. The experimental setup of XJTUSpurgear Dataset.

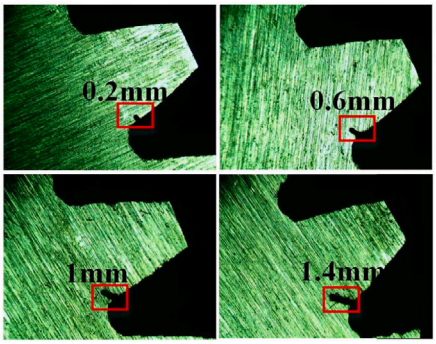


Fig. 15. Health conditions of spur-gears.

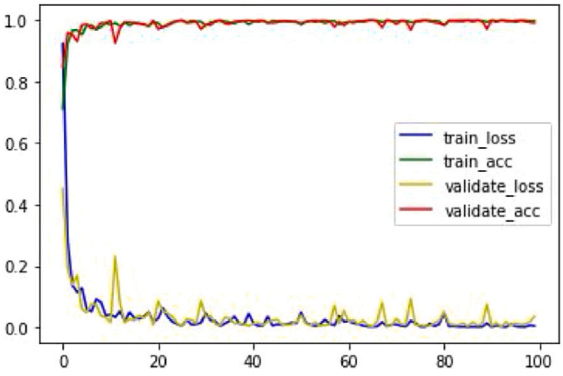


Fig. 16. Plot of accuracy and loss rate versus the number of training rounds of the model under XJTUSuprgear gearbox dataset.

both training and validation datasets, and exhibits robust generalization capability. As shown in the results of the confusion matrix in Fig. 17, the different broken teeth in the XJTUSuprgear dataset show good diagnostic results, with only a small number of misclassifications occurring when the broken teeth are 1 mm. These data demonstrate the accuracy and versatility of the fault diagnosis research methodology proposed in this paper, which can accurately and efficiently identify the faults of different components in wind turbines under variable operating conditions.

As shown in Fig. 18, one can observe the feature distribution of the model both prior to and subsequent to training. This visual representation offers an intuitive and comprehensive insight into the model's data classification capabilities. In the initial distribution depicted in Fig. 18(a), data belonging to diverse categories appears to be in a state of disarray. However, as the training of the model progresses and deepens, the output distribution presented in Fig. 18(b) reveals a notable transformation. Here, a distinct separation between different classes becomes evident, while data points within the same class tend

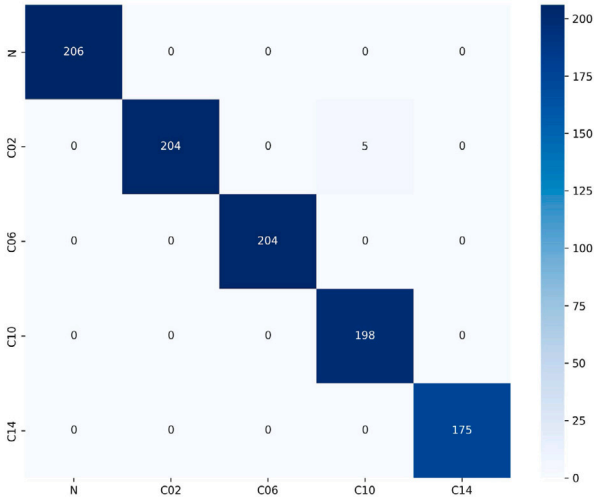
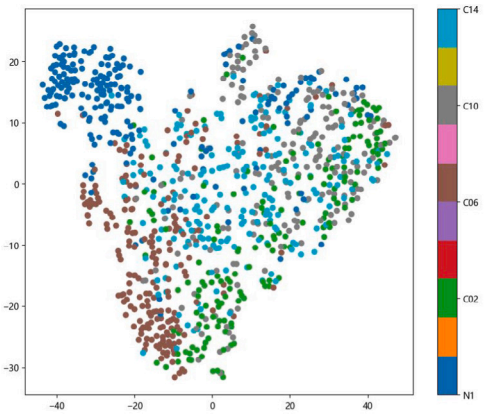
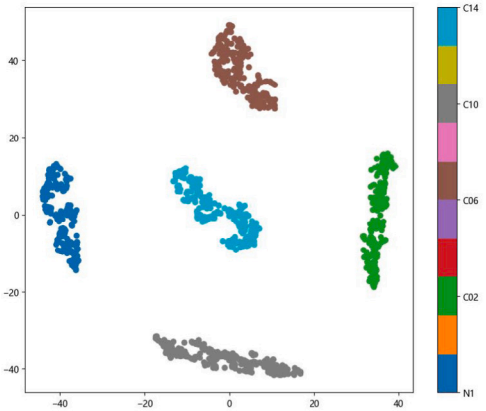


Fig. 17. Confusion matrix at variable speed XJTUSuprgear gearbox.



(a) Initial distribution.



(b) Output distribution.

Fig. 18. Visualisation of the model training process.

to cluster together in a relatively cohesive manner. By comparing these two distributions, it becomes clear that the model proposed in this research demonstrates a robust capacity to distinguish fault-type data from fault-free data.

Grad-CAM++ (Gradient-weighted Class Activation Mapping) is a widely used visualisation method for highlighting key areas by calculating the gradient of a target category to a feature map layer, combining

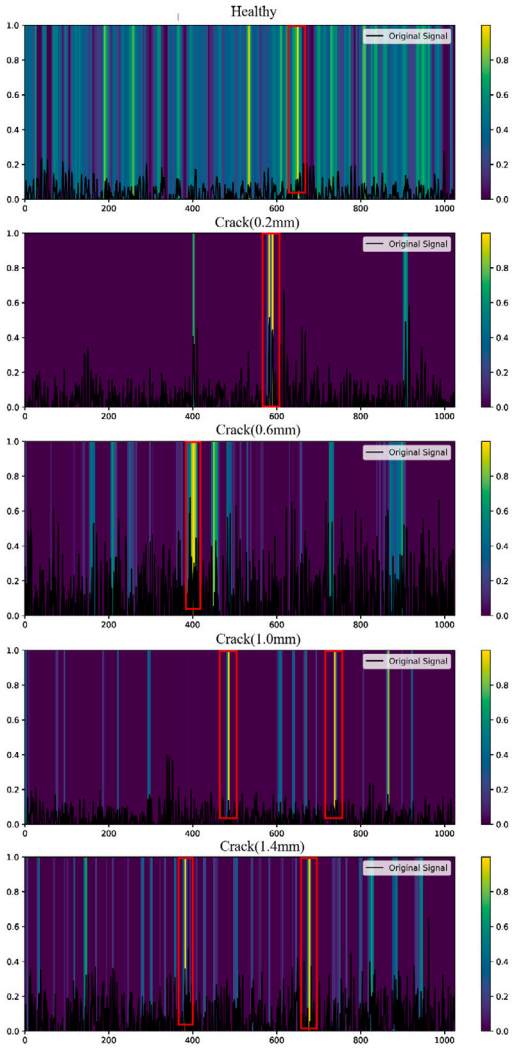


Fig. 19. Grad-CAM++ visualisation of five fault states.

higher order gradient information to generate weights, and weighting the feature map to produce a heatmap.

To further investigate the interpretability of the method, the Class Activation Mapping (CAM) is calculated by extracting the feature map of the model convolution kernel, as shown in Fig. 19. The colour scale intensity is positively correlated with the CAM value, i.e. the higher the colour scale, the higher the corresponding CAM value and the more significant the feature differentiation, and the red boxed area in the figure is the part of the high CAM value. It can be seen that there are differences in CAM activation position and amplitude for different flaw types, which fully reflects the characteristics of TFNet to achieve flaw type differentiation without manual pre-processing. For example, crack (1.0 mm) and crack (1.4 mm) show different activation areas; for different damage levels of flake samples, there are also differences in the activation areas, and as the damage level deepens, the impact amplitude becomes more obvious.

To be more convincing, we conducted comparative experiments with the seven models in Section 4. The results of the different models in the test set after 100 iterations of training for each evaluation metric are presented in Table 7. It can be seen that the TFNet model achieves 98.84% in accuracy, precision, recall and F1 score with a training time of 956 s. Compared to other models, TFNet has a clear advantage in diagnostic performance indicators and although the training time is relatively long compared to some models, overall it shows higher

Table 7
Comparison of evaluation indicators results for different model test sets.

Model	Accuracy (%)	Precision (%)	Recall (%)	F1_score (%)	Training times (s)
TFNet	98.84	98.85	98.84	98.84	956
CNN	91.50	91.50	91.55	91.55	478
WDCNN	94.25	94.20	94.25	94.25	506
Resnet	95.60	95.53	95.55	95.55	893
LSTM	96.88	96.90	96.90	96.88	467
MLFNet	97.54	97.55	97.50	97.55	1024
MSACNN	97.55	97.68	97.60	97.31	1320
MSECTN	98.10	98.04	98.32	98.22	1267

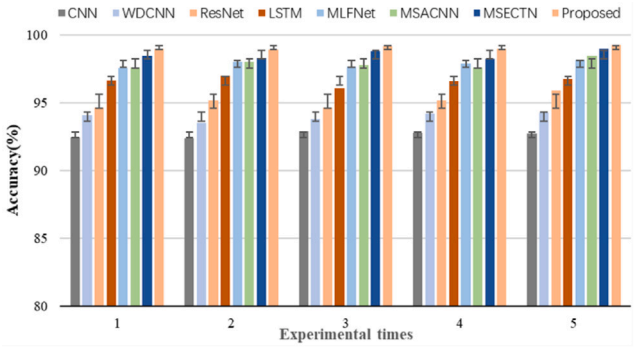


Fig. 20. Results of different methods for XJTUSuprgear gearbox dataset.

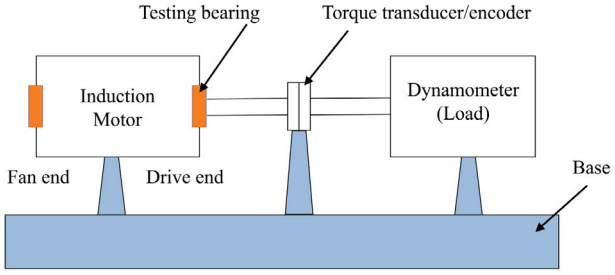


Fig. 21. CWRU experimental device platform.

reliability and validity on the test set and performs well in the fault diagnosis task. Five randomised trials were conducted to enhance the reliability and generalisation of the results. The experimental results are shown in Fig. 20. Obviously, the developed TFNet diagnosis is the most effective, with test results precision of 99.70%, 99.65%, 99.91%, 99.50% and 99.98% for the five tests, accordingly. TFNet's best accuracy is superior to the best of the following seven methods. In addition, TFNet has the lowest standard deviation compared to the other methods.

5.2. Fault identification of CWRU bearing dataset

Bearing failure data from Case Western Reserve University [45] (CWRU) is used to verify the generality of the model for bearing failure diagnosis. As shown in Fig. 21, the main components of the fault test bench are an induction motor, a torque sensor and a power meter. The CWRU data set is used to collect vibration signals from the drive end bearing under four different operating conditions sampling frequency was 12 kHz. The bearing type is SKF6205. The four operating conditions are Normal, Inner Race Failure, Ball Failure and Outer Race Failure.

The collected data are categorised into 10 labels according to failure types, and each failure type contains 200 training data divided into training set, validation set and test set in the ratio of 6:2:2. The experimental data set is divided as shown in Table 8.

Table 8
Sample status of rolling bearing fault data.

Health condition	Fault size (inches)	Fault Type	Sample size	Class labels
N	–	–	200	N
IR	0.007	IR007	200	IR07
	0.014	IR014	200	IR14
	0.021	IR021	200	IR21
OR	0.007	OR007	200	OR07
	0.014	OR014	500	OR14
	0.021	OR021	500	OR21
BF	0.007	BO007	200	B07
	0.014	BO014	200	B14
	0.021	BO021	200	B21

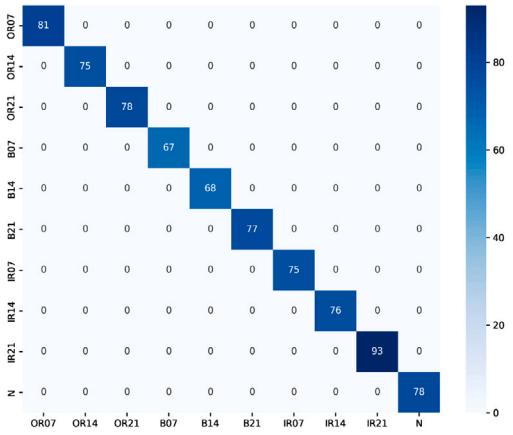


Fig. 22. Confusion matrix at CWRU.

To further validate the generalisation performance of the proposed method on the bearing dataset, confusion matrix experiments are performed and the results are shown in Fig. 22. From the confusion matrix, it can be seen that the model correctly predicts the categories in all 10 faults. This indicates that the proposed method has high fault identification performance and good generalisation performance on the bearing dataset. Again, as shown in Fig. 23, the visualisation of the extracted features of the model shows that the hidden features extracted by TFNet can accurately separate all the faults.

Based on the comparison of the evaluation metrics of different model test sets under the CWRU data in Table 9, it can be seen that the TFNet model shows excellent performance in the main evaluation metrics of accuracy, precision, recall and F1 score, which are 99.64%, 99.55%, 99.64% and 99.60% respectively, and outperforms CNN, WDCNN and other comparative models, Resnet, LSTM and other comparative models. Although the training time of TFNet is 1026 s, which is relatively long compared to some models, it has a significant advantage in fault diagnosis accuracy, indicating that TFNet has high reliability and effectiveness in fault diagnosis using CWRU data.

Table 9
Comparison of evaluation metrics results for different model test sets under CWRU data.

Model	Accuracy (%)	Precision (%)	Recall (%)	F1_score (%)	Training times (s)
TFNet	99.64	99.55	99.64	99.60	1026
CNN	94.80	94.80	94.70	94.78	528
WDCNN	95.66	95.60	95.60	95.70	596
Resnet	96.80	96.78	96.83	96.78	887
LSTM	97.78	97.77	97.80	97.88	567
MLFNet	98.43	98.55	98.55	98.55	1154
MSACNN	98.75	98.68	98.60	98.81	1420
MSECTN	99.10	99.04	99.12	99.18	1369

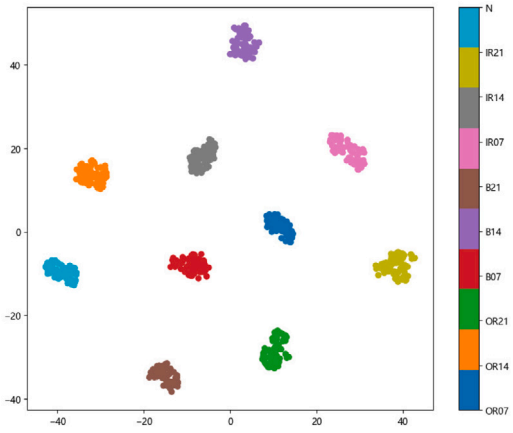


Fig. 23. T-SNE feature visualisation for features under TFNet models.



(a) Roller press and bearing failure rate of roller press.



(b) Centrifugal fan and centrifugal fan bearing failure.

Fig. 24. Cement production equipment of a cement plant.

6. Engineering validation

In practical engineering applications, the TFNet model is used to identify mechanical equipment faults in complex environments, specifically in the cement production environment for performance verification of a variety of production equipment. The production environment is very harsh and filled with a lot of noise interference, which poses a greater challenge to the model's crucial fault feature extract and fault identification accuracy.

The cement production equipment selected included roller presses and centrifugal fans, each of which suffered from different types of faults: internal raceway faults in roller presses and imbalance faults in centrifugal fans. As shown in Fig. 24, system sampling frequency is 51.2 kHz, sampling time is 2 s, and sampling points are 102.4k. Unbalance fault refers to the large rotating equipment, rotor by the material, quality, processing, assembly and operation of a variety of factors, the centre of mass and the centre line of rotation between the existence of a certain degree of eccentricity phenomenon, so that the rotor in the work of the formation of cyclical centrifugal force interference, and ultimately lead to mechanical vibration or even lead to mechanical equipment stoppage and destruction of the phenomenon. Table 10 shows the distribution of the number of defect samples and the corresponding labels for each device.

Table 10
Data set segmentation and corresponding labelling.

Equipment name	Fault type	Sample size	Class labels
Roller press	Normal	500	N
	Fault	500	F
Centrifugal fan	Normal	500	N
	Fault	500	F

The Fig. 25 confusion matrix shows the diagnostic outputs of the model in two types of cement plant equipment failure data. The models can be seen to correctly predict categories in both roller press and centrifugal fan equipment failures, confirming the powerful classification performance of the models.

Based on a specific dataset of cement plant production equipment, this study validates the fault diagnosis accuracy of the TFNet model and compares it with similar deep learning models in Section 4 to comprehensively consider their performance differences in equipment fault identification. To ensure the accuracy of the test results, five independent tests are conducted for each experiment. Fig. 26 illustrates the accuracy distribution characteristics of different models in the fault detection of cement production equipment groups. It is worth noting that the TFNet model has the smallest standard deviation of accuracy rate on each equipment type, which verifies the excellent and comprehensive intelligent diagnosis performance and superior stability of the model under complex working conditions. It further confirms the effectiveness of its unique noise immunity mechanism in industrial noise environments.

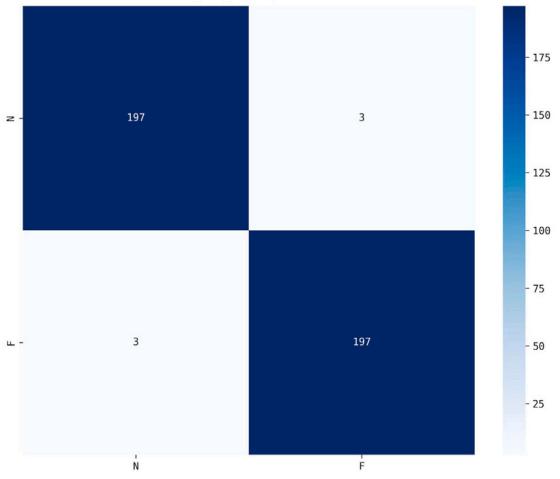
Table 11 compares the training time accuracy of the TFNet model with other models. The results show that the training time of the proposed model is faster than the other models except CNN, WDCNN and LSTM for both cement production plants. Overall, although the training time of TFNet is not the shortest in some of the comparisons, it is able to complete the training with a relatively reasonable training time compared to other complex models for the diagnostic task of cement production equipment, which shows better time efficiency and adaptability in practical applications.

7. Conclusions

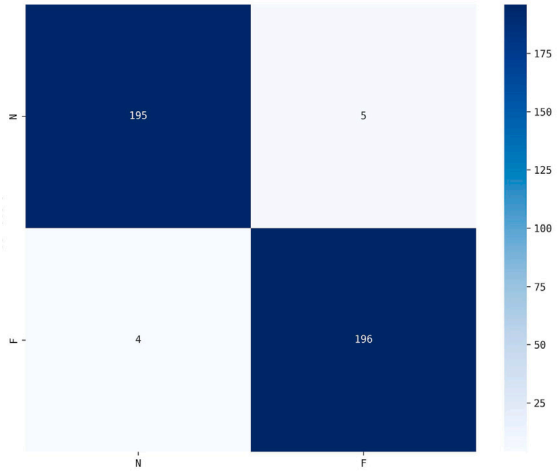
In this paper, we propose a method called TFNet for mechanical fault diagnosis of noisy wind turbines, which uses different methods to extract information about different fault features from multiple feature domains. The proposed method constructs a time–frequency ensemble network which contains two single branches (T-Block and F-Block) to extract multi-domain features. And a label smoothing algorithm is used to assist the model training. To evaluate the performance of TFNet, a case study of the WT-Planetary gearbox dataset was implemented. The

Table 11
Diagnosis performance of different methods on the cement production equipment.

Model	Equipment name	Training time (s)	Equipment name	Training time (s)
TFNet	Roller press	389	Centrifugal fan	405
CNN		284		286
WDCNN		341		334
Resnet		568		588
LSTM		352		362
MLFNet		785		798
MSACNN		745		822
MSECTN		912		895



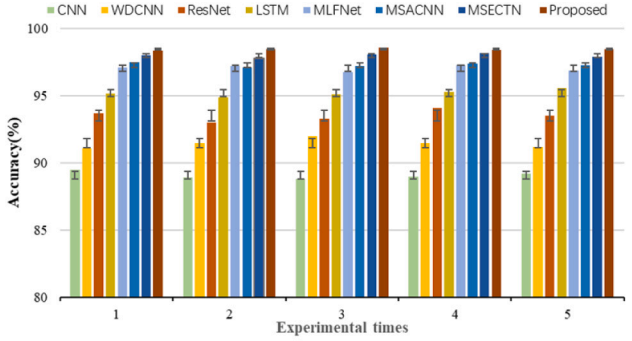
(a) Roller press.



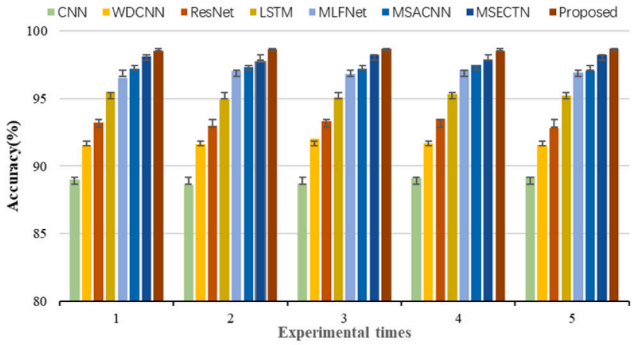
(b) Centrifugal fan.

Fig. 25. TFNet confusion matrix on cement production equipment.

utility of the model was verified using data from the XJTUSuprgear variable speed gearbox dataset. Testing the TFNet model for faults in other components using the CWRU bearing dataset. In addition, specific engineering validation was carried out using two different data sets in a cement production process environment in a cement plant. Comprehensive results show that the proposed method achieves the highest average test accuracy and is more stable compared to other methods. Especially under noisy conditions. We also carried out ablation studies to further validate the performance of ASB, T-Block and F-Block. In future work, we will apply TFNet to multimodal data and multi-sensor data fusion.



(a) Roller press.



(b) Centrifugal fan.

Fig. 26. Accuracy of different models with different cement equipment.

CRediT authorship contribution statement

Haiyu Guo: Writing – review & editing. **Xingzheng Guo:** Writing – original draft. **Xiaoguang Zhang:** Investigation. **Fanfan Lu:** Software. **Chuang Liang:** Software.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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References

[1] R. Zhou, Y. Li, X. Lin, A clustered federated learning framework for collaborative fault diagnosis of wind turbines, Appl. Energy 377 (2025) 124532.

[2] W. Zhang, J. Vatn, A. Rasheed, Gearbox pump failure prognostics in offshore wind turbine by an integrated data-driven model, Appl. Energy 380 (2025) 124829.
[3] H. Guo, W. Yu, X. Zhang, F. Lu, C. Liang, Intelligent mechanical fault diagnosis using multiscale residual network and multisensor fusion, Meas. Sci. Technol. 35 (11) (2024) 116007.
[4] M. Tang, C. Cao, H. Wu, H. Zhu, J. Tang, Z. Peng, Y. Wang, Fault detection of wind turbine gearboxes based on IBOA-ERF, Sensors 22 (18) (2022) 6826.
[5] F. Bie, X. Ding, Q. Li, Y. Zhang, X. Huang, A fault diagnosis method for planetary gearboxes based on IFMD, Shock. Vib. 2024 (1) (2024) 2140227.
[6] R.G. Desavale, P.M. Jadhav, N.V. Dharwadkar, Dynamic response analysis of gearbox to improve fault detection using empirical mode decomposition and artificial neural network techniques, ASCE- ASME J. Risk Uncertain. Eng. Syst. Part B: Mech. Eng. 7 (3) (2021) 031007.
[7] X. Liu, G. Wang, G. Wu, X. Li, Fault diagnosis method of planetary gear box based on variational modal decomposition and particle swarm optimization support vector machine, J. Electron. Meas. Instrum. 36 (1) (2023) 54–61.
[8] C. Liang, X. Mu, X. Zhang, F. Lu, C. Wang, Y. Shao, Enhanced fault diagnosis of rolling bearings using attention-augmented separable residual networks, Eng. Sci. Technol. Int. J. 61 (2025) 101930.
[9] Z. Zhu, Y. Lei, G. Qi, Y. Chai, N. Mazur, Y. An, X. Huang, A review of the application of deep learning in intelligent fault diagnosis of rotating machinery, Meas. 206 (2023) 112346.
[10] X. Yin, Z. Mou, Y. Wang, Fault diagnosis of wind turbine gearbox based on multiscale residual features and ECA-stacked ResNet, IEEE Sensors J. 23 (7) (2023) 7320–7333.
[11] Y. Wang, J. Wang, P. Tong, Small sample fault diagnosis for wind turbine gearbox based on lightweight multiscale convolutional neural network, Meas. Sci. Technol. 34 (9) (2023) 095111.
[12] Q. Yang, B. Tang, Y. Shen, Q. Li, Self-attention parallel fusion network for wind turbine gearboxes fault diagnosis, IEEE Sensors J. (2023).
[13] Y. Chen, X. Liu, M. Rao, Y. Qin, Z. Wang, Y. Ji, Explicit speed-integrated LSTM network for non-stationary gearbox vibration representation and fault detection under varying speed conditions, Reliab. Eng. Syst. Saf. 254 (2025) 110596.
[14] D. Zhao, D. Shao, L. Cui, CTNet: A data-driven time-frequency technique for wind turbines fault diagnosis under time-varying speeds, ISA Trans. 154 (2024) 335–351.
[15] S. Munikoti, D. Agarwal, L. Das, M. Halappanavar, B. Natarajan, Challenges and opportunities in deep reinforcement learning with graph neural networks: A comprehensive review of algorithms and applications, IEEE Trans. Neural Netw. Learn. Syst. (2023).
[16] W. Xiang, S. Liu, H. Li, S. Cao, K. Zhang, C. Yang, Fault diagnosis of gearbox based on refined topology and spatio-temporal graph convolutional networks, IEEE Sensors J. (2023).
[17] P. Yin, J. Nie, X. Liang, S. Yu, C. Wang, W. Nie, X. Ding, A multi-scale graph convolutional neural network framework for fault diagnosis of rolling bearing, IEEE Trans. Instrum. Meas. (2023).
[18] D. Zhao, W. Cai, L. Cui, Multi-perception graph convolutional tree-embedded network for aero-engine bearing health monitoring with unbalanced data, Reliab. Eng. Syst. Saf. (2025) 110888.
[19] A. Vaswani, Attention is all you need, Adv. Neural Inf. Process. Syst. (2017).
[20] Y. Zhang, Y. Tang, Y. Liu, Z. Liang, Fault diagnosis of transformer using artificial intelligence: A review, Front. Energy Res. 10 (2022) 1006474.
[21] K. Guan, B. Du, Z. Wu, J. Li, Y. Zhang, Fault diagnosis of gearbox based on improved transformer, in: Proceedings of the 2023 International Conference on Advances in Artificial Intelligence and Applications, 2023, pp. 312–318.
[22] S. Han, H. Shao, J. Cheng, X. Yang, B. Cai, Convformer-NSE: A novel end-to-end gearbox fault diagnosis framework under heavy noise using joint global and local information, IEEE/ASME Trans. Mechatronics 28 (1) (2022) 340–349.
[23] Y. Zhang, X. Zhao, H. Liang, P. Chen, Multiscale dilated convolution and swin-transformer for small sample gearbox fault diagnosis, Appl. Intell. (2024) 1–17.
[24] Y. Xiao, H. Shao, J. Wang, S. Yan, B. Liu, Bayesian variational transformer: A generalizable model for rotating machinery fault diagnosis, Mech. Syst. Signal Process. 207 (2024) 110936.
[25] Y. Xu, Y. Chen, H. Zhang, K. Feng, Y. Wang, C. Yang, Q. Ni, Global contextual feature aggregation networks with multiscale attention mechanism for mechanical fault diagnosis under non-stationary conditions, Mech. Syst. Signal Process. 203 (2023) 110724.
[26] Z. Zhao, T. Li, B. An, S. Wang, B. Ding, R. Yan, X. Chen, Model-driven deep unrolling: Towards interpretable deep learning against noise attacks for intelligent fault diagnosis, ISA Trans. 129 (2022) 644–662.
[27] B. An, S. Wang, Z. Zhao, F. Qin, R. Yan, X. Chen, Interpretable neural network via algorithm unrolling for mechanical fault diagnosis, IEEE Trans. Instrum. Meas. 71 (2022) 1–11.
[28] B. An, S. Wang, F. Qin, Z. Zhao, R. Yan, X. Chen, Adversarial algorithm unrolling network for interpretable mechanical anomaly detection, IEEE Trans. Neural Netw. Learn. Syst. (2023).
[29] Y. Ding, M. Jia, Q. Miao, Y. Cao, A novel time–frequency transformer based on self–attention mechanism and its application in fault diagnosis of rolling bearings, Mech. Syst. Signal Process. 168 (2022) 108616.
[30] X. Feng, G. Zhao, Y. Wang, D. Gao, H. Ding, Fault diagnosis method based on the multi-head attention focusing on data positional information, Meas. Control. 56 (3–4) (2023) 583–595.
[31] E. Eldele, M. Ragab, Z. Chen, M. Wu, X. Li, Tslanet: Rethinking transformers for time series representation learning, 2024, arXiv preprint arXiv:2404.08472.
[32] J. Hua, Y. Zhang, D. Zhang, J. He, J. Wang, X. Fang, Dynamic feature-aware graph convolutional network with multi-sensor for remaining useful life prediction of Turbofan engines, IEEE Sensors J. (2024).
[33] C. Szegedy, V. Vanhoucke, S. Ioffe, J. Shlens, Z. Wojna, Rethinking the inception architecture for computer vision, in: Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition, 2016, pp. 2818–2826.
[34] D. Liu, L. Cui, W. Cheng, A review on deep learning in planetary gearbox health state recognition: Methods, applications, and dataset publication, Meas. Sci. Technol. (2023).
[35] J. Wang, C. Zhan, D. Yu, Q. Zhao, Z. Xie, Rolling bearing fault diagnosis method based on SSAE and softmax classifier with improved K-fold cross-validation, Meas. Sci. Technol. 33 (10) (2022) 105110.
[36] D. Ruan, J. Wang, J. Yan, C. Gühmann, CNN parameter design based on fault signal analysis and its application in bearing fault diagnosis, Adv. Eng. Inform. 55 (2023) 101877.
[37] D. Lee, J. Jeong, Few-shot learning-based light-weight WDCNN model for bearing fault diagnosis in siamese network, Sensors 23 (14) (2023) 6587.
[38] C. He, Y. Cao, Y. Yang, Y. Liu, X. Liu, Z. Cao, Fault diagnosis of rotating machinery based on the improved multidimensional normalization ResNet, IEEE Trans. Instrum. Meas. (2023).
[39] D. Gao, Y. Zhu, Z. Ren, K. Yan, W. Kang, A novel weak fault diagnosis method for rolling bearings based on LSTM considering quasi-periodicity, Knowl.-Based Syst. 231 (2021) 107413.
[40] Z. Ye, J. Yu, Multi-level features fusion network-based feature learning for machinery fault diagnosis, Appl. Soft Comput. 122 (2022) 108900.
[41] L. Deng, Y. Zhang, Z. Shi, An improved fault diagnosis method of rolling bearings based on multi-scale attention CNN, J. Fail. Anal. Prev. 24 (4) (2024) 1814–1827.
[42] S. Wang, Z. Liu, Z. Jia, W. Zhao, Z. Li, Intermittent fault diagnosis for electronics-rich analog circuit systems based on multi-scale enhanced convolution transformer network with novel token fusion strategy, Expert Syst. Appl. 238 (2024) 121964.
[43] Y. Ji, Y. Huang, J. Zeng, L. Ren, Y. Chen, A physical–data-driven combined strategy for load identification of tire type rail transit vehicle, Reliab. Eng. Syst. Saf. 253 (2025) 110493.
[44] T. Li, Z. Zhou, S. Li, C. Sun, R. Yan, X. Chen, The emerging graph neural networks for intelligent fault diagnostics and prognostics: A guideline and a benchmark study, Mech. Syst. Signal Process. 168 (2022) 108653.
[45] M. Alonso-González, V.G. Díaz, B.L. Pérez, B.C.P. G-Bustelo, J.P. Anzola, Bearing fault diagnosis with envelope analysis and machine learning approaches using CWRU dataset, IEEE Access 11 (2023) 57796–57805.